

Lecture-4

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Functions of Control Unit of 8086:

- **Data bus :**

Carries data in binary form between microprocessor and other external units and vice-versa. The data bus width of 8086 microprocessor is 16-bit i.e. 2^{16} combination of binary digits and are typically identified as D0 – D15. So the largest number is 65536 in decimal.

- **Address Bus:**

It contains 20 bit address bus and identified as A0-A19.

- **(AD0 –AD15) [Address/data bus]:** address bus are multiplexed with (D0-D15) data bus.
- **(A16/S3 –A19/S6) [Address/status bus]:** there are 4 address/status bus. During the first clock cycle it carries 4bit address, later it carries status signal.

- **Control bus:**

Carries control signals. The following control and status signals used in 8086 microprocessor.

Pin in max mode-

- $\overline{\text{BHE}}/\text{S}_7$ - bus high enable is an active low signal, so it active when $\text{BHE} = 0$.
- $\overline{\text{RD}}$ - this is used for read operation when $\overline{\text{RD}} = 0$.
- $\text{MN}/\overline{\text{MX}}$ -

{	1, maximum mode
	0, minimum mode

- $(\overline{S_2}, \overline{S_1}, \overline{S_0})$ – status signal and associated operation.

Status Inputs			CPU Cycles
$\overline{S_2}$	$\overline{S_1}$	$\overline{S_0}$	
0	0	0	Interrupt Acknowledge
0	0	1	Read I/O Port
0	1	0	Write I/O Port
0	1	1	Halt
1	0	0	Instruction Fetch
1	0	1	Read Memory
1	1	0	Write Memory
1	1	1	Passive

Pin in min mode-

- **WR-** 0, then write operation active.
- **ALE-** address appears on (AD0-AD15).
- **$\overline{\text{IO/M}}$ -** $\left\{ \begin{array}{l} 0, \text{ I/O operation} \\ 1, \text{ memory operation} \end{array} \right.$